* ANALYTICAL PROBLEM - SOLVING ACTIVITY. *

1. Gradient magnitude and Gradient Direction:

Given:

- $G_x = 24$
- $G_y = 18$

∴ Gradient magnitude.

Formula:

$$G = \sqrt{G_x^2 + G_y^2}$$

$$G = \sqrt{24^2 + 18^2} = \sqrt{576 + 324}$$

$$G = \sqrt{900} \Rightarrow \underline{G = 30}$$

· Gradient Direction:

$$\theta = \tan^{-1}(G_y / G_x)$$

$$= \tan^{-1}(18/24)$$

$$\theta = \tan^{-1}(0.75) = 36.87°$$

∴ Gradient magnitude $= 30$

Gradient Direction $= 36.87°$.

2. Non-maximum suppression:

Given gradient magnitudes:

Pixel	P1	P2	P3	P4	P5
magnitude	25	50	90	60	30

· Gradient direction is horizontal, so each pixel is compared with its left and right neighbors.

- P1 = 25 → Less than P2(50) → Suppress
- P2 = 50 → Less than P3 (90) → Suppress
- P3 = 90 → greater than P2(50) & P4(60) → Keep
- P4 = 60 → Less than P3 (90) → Suppress
- P5 = 30 → Less than P4 (60) → Suppress

NMS Result:

Pixel	P1	P2	P3	P4	P5
After NMS	0	0	90	0	0

∴ P3 = 90 is the only remaining edge pixel.

3. Double thresholding:

given;

- High Threshold = 100
- Low Threshold = 50

Pixel	magnitude	classification.
P1	135	Strong edge
P2	90	weak
P3	45	Non-edge
P4	120	strong
P5	70	weak
P6	20	Non-edge

Rule:

- magnitude ≥ 100 → strong-edge
- 50 - 99 → weak-edge
- < 50 → Non-edge

4. Edge Tracking by Hysteresis:

1. Strong-edges:

P_1, P_4, P_7

2. Weak-Edges:

Initially:- P_2, P_5

3. Non-Edges:

P_3, P_6.

4. Final Edge pixels After Hysteresis:

- P1 → Strong → Keep
- P2 → Connected to P1 → Keep
- P3 → Non-edge → Remove
- P4 → Strong → Keep
- P5 → Not connected to Strong → Remove
- P6 → Non-edge → Remove
- P7 → Strong → Keep.

∴ Final Answer: P1, P2, P4, P7